

Muhammad Azam

AI Automation Engineer

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SUMMARY

AI Automation Engineer who designs and ships production AI systems end to end -- multi-agent LLM pipelines grounded in real data, full-stack platforms with real payments and video, and the CI/CD that keeps them reliable. Four flagship products are live today, not prototypes.

COMPLETED PROJECTS

CoolCore -- coolcore.vercel.app

- Heat-exposure decision-support platform built in 7 days for the FortyGuard Hackathon '26 (Team Coolcore); live on Next.js 16 + Vercel, integrating FortyGuard's real heat-index/wet-bulb API.
- Shipped a shift-exposure timeline, a shade-scored transit route comparator using streetview image segmentation, and a supervisor-facing outdoor-worker safety board.

QuranicLens -- quraniclens.vercel.app

- 4-agent LLM pipeline (DeepSeek, Claude Sonnet/Opus, Gemini) synthesizes grounded answers across 6 lenses -- tafsir, hadith, science, history, and scripture parallels.
- 78,000+ records embedded with pgvector (Postgres/Supabase); retrieval-first architecture forbids the model from citing anything outside verified source data.

FreshTrack -- freshtrack-sigma.vercel.app

- Mobile-first PWA for restaurant inventory: batch-level FEFO tracking, barcode scanning, and rule-based reorder forecasting.
- Multi-tenant access control (Supabase Auth), CSV import, and waste analytics dashboards built on Drizzle ORM + Postgres.

LMS Platform -- [self-hosted on AWS EC2 | github.com/MAzam1999/Devweekends](https://self-hosted-on-AWS-EC2-github.com/MAzam1999/Devweekends)

- Dev Weekends Fellowship project: course marketplace with Stripe Checkout, Mux signed video playback, and Clerk role-based access (student/instructor/admin).
- One codebase ships to production two ways -- Vercel, and a self-hosted Docker container on EC2 via a GitHub Actions -> ECR -> EC2 pipeline.

Portfolio Site with Case Studies -- portfolio-mazam1999s-projects.vercel.app

- Dev Weekends Fellowship project: dependency-free static site (HTML/CSS/JS) on Vercel with three deep-dive case studies (QuranicLens, FreshTrack, LMS Platform).

EXPERIENCE

AI Automation Engineer -- Independent

2024 -- Present, Remote

- Designed and shipped three production AI/full-stack products end to end: architecture, data model, AI pipeline, deployment.
- Built multi-agent LLM pipelines using retrieval-grounded context instead of model training data.

Fellow -- Dev Weekends Fellowship 2026

Summer 2026

- Completed the Engineering L0 & L1, DSA, Data Engineering, and Cloud & DevOps class series; delivered the LMS Platform and portfolio projects above.

Data Science Intern -- The Spark Foundation

09/2023 -- 10/2023, Remote

- Designed and optimized machine learning models to improve prediction accuracy; built data visualizations for actionable insight.

Data Science Intern -- Omdena

08/2023 -- 09/2023, Remote

- Cleaned and preprocessed large datasets for a fake-news detection model; ran EDA for a weather-forecasting initiative with a global team. Recommended in writing by Omdena's Community Manager.

Workflow Automation Specialist -- Freelance

12/2019 -- 01/2023, Remote

- Designed end-to-end workflow automation and integrated AI tools into client systems for data collection, analysis, and reporting.

SKILLS

Languages and Frameworks: TypeScript, JavaScript, Python, Next.js 16, React 19

Data and AI: PostgreSQL, pgvector, Prisma, Drizzle ORM, multi-agent LLM orchestration, RAG, OpenRouter, embeddings, data engineering

Infra and DevOps: Docker, AWS (EC2, ECR), GitHub Actions CI/CD, Vercel, nginx

Integrations and Platform: Stripe, Clerk, Mux, Supabase, NextAuth, n8n

Computer Science: Data Structures and Algorithms, competitive programming

CERTIFICATIONS AND ACHIEVEMENTS

- **Dev Weekends Fellowship 2026 (Summer)** -- Certificate of Participation, 09/2026
- **FortyGuard Hackathon '26** -- Team Coolcore; built and shipped CoolCore
- **Meta Hacker Cup 2023** -- Advanced to Round 1 (Practice Round rank 2,590; Round 1 rank 6,262)
- **Advent of Code 2023** -- Completed
- **Stanford Code in Place 2024** -- Certificate of Completion, all units and course exam
- **Aspire Leaders Program 2025** -- Module 1: Personal and Professional Development (15 hours)
- **Omdena** -- 2 Local Chapter product projects completed; Rookie Collaborator badge; letter of recommendation from Omdena
- **Coursera** -- Machine Learning Specialization; Neural Networks and Deep Learning; Generative AI with LLMs; AI Python for Beginners (DeepLearning.AI) | Statistics with Python; Programming for Everybody (University of Michigan)

EDUCATION

Bachelor of Computer Science (Expected) -- Virtual University of Pakistan, since 02/2025

LANGUAGES

Urdu -- Native | **English** -- Professional Working Proficiency